

Experiment No. 7

Aim –

Perform DOM traversal and updates; develop a To-Do List application with add, edit, and delete features. Implement a registration form using JavaScript event handling and form validation.

Tools Required –

- VS Code
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

Theory –

1) Document Object Model (DOM)

The Document Object Model (DOM) represents an HTML document as a tree of objects.

JavaScript uses the DOM to access, modify, create, and remove HTML elements dynamically.

2) DOM Traversal and Manipulation

DOM manipulation allows JavaScript to dynamically update webpage content.

Common

methods used include:

- getElementById() • createElement()
- appendChild()
- removeChild()
- querySelectorAll()

These methods are used to create, add, update, and remove elements from a webpage.

3) Event Handling

Events are actions performed by the user, such as clicking a button, entering text, changing a

selection, or submitting a form.

In this experiment, the following events are used:

- click – for adding, editing, and deleting tasks
- input – for real-time validation
- change – for selection and checkbox validation
- submit – for validating the complete form

4) To-Do List Application

The To-Do List application allows users to manage tasks dynamically. A user can add a new

task, edit an existing task, or delete an unwanted task. JavaScript creates and updates the required elements without reloading the webpage.

5) Form Validation

Form validation checks whether the user has entered correct information before submission.

The registration form validates required fields, username, email, website, password, birthday, and terms and conditions. Error messages are displayed dynamically for invalid input.

6) Regular Expressions

Regular Expressions (Regex) are patterns used to check whether text follows a required format.

They are used in this experiment to validate the email address, website address, and password requirements.

Conclusion –

The experiment was successfully performed to understand DOM traversal, manipulation, event handling, and form validation using JavaScript. A To-Do List application with Add, Edit, and Delete features was developed, and a registration form was implemented with dynamic validation and error handling.